On the other hand, the data layer requires some extra work to address two main concerns:

- How can we tell which users created what data? Who changed it last?
- How can we ensure that each user can only access their own book orders?

This section will address both concerns. First I'll explain how to enable auditing for users' actions on data in both Catalog Service and Order Service. Then I'll walk you through the changes required by Order Service to keep the data private.

12.4.1 Auditing data with Spring Security and Spring Data JDBC

Let's start by considering Catalog Service, where the data layer is implemented with Spring Data JDBC. In chapter 5, you learned how to enable JDBC data auditing, and you configured it to save both the creation date and last modified date for each data entity. Building on top of that, we can now extend the audit scope to include the usernames of the person who created the entity and the person who modified it last.

First we need to tell Spring Data where to get the information about the currently authenticated user. In the previous chapter, you learned that Spring Security stores information about authenticated users in an `Authentication` object, which is stored in a `SecurityContext` object available through the `SecurityContextHolder`. We can use that object hierarchy to specify how to extract the principal for Spring Data.